M.A.P.A.S(MULTIPURPOSE AUTONOMOUS AND PILOTED AIRCRAFT SYSTEM)

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***Abstract*—The Multipurpose Autonomous and Piloted Aircraft System (M.A.P.A.S) presents a hybrid dual-drone architecture designed to address the operational limitations of conventional single-role unmanned aerial vehicles. The proposed system integrates a long-endurance autonomous carrier drone with a compact, manually piloted first-person-view (FPV) drone to achieve both wide-area coverage and high-precision localized intervention. Autonomous navigation and mission execution are supported through onboard computing, sensor fusion, and computer vision techniques, enabling real-time object and human detection for emergency response, surveillance, and resource delivery applications. A modular payload framework allows rapid reconfiguration for tasks such as agricultural spraying, rescue equipment deployment, public address communication, and aid distribution. The piloted sub-drone is deployed when close-range maneuverability is required in constrained or hazardous environments. By combining autonomous decision-making with human-in-the-loop control, the proposed system improves mission flexibility, reduces response time, and enhances operational safety. The M.A.P.A.S platform demonstrates a scalable and cost-effective solution for disaster management, public safety, environmental monitoring, and humanitarian operations.**

***Index Terms*—Autonomous drones, dual-drone system, unmanned aerial vehicles (UAVs), computer vision, modular payload, disaster response, FPV drone, human-in-the-loop control, aerial surveillance, humanitarian applications**

I. Introduction

Unmanned Aerial Vehicles (UAVs) are increasingly used in applications such as disaster response, agriculture, surveillance, and environmental monitoring due to their rapid deployment and ability to operate in hazardous or inaccessible environments. However, most existing UAV platforms are designed for single-purpose missions, which limits operational flexibility and increases deployment cost when multiple tasks or precision interventions are required.

This paper presents the Multipurpose Autonomous and Piloted Aircraft System (M.A.P.A.S), a hybrid dual-drone platform that integrates autonomous wide-area operation with human-in-the-loop precision control. The system combines a long-endurance autonomous carrier drone with a compact FPV-based piloted drone, supported by computer vision and a modular payload architecture. By enabling coordinated autonomous and piloted missions within a unified framework, M.A.P.A.S improves mission adaptability, response efficiency, and operational safety for applications including disaster management, public safety, and humanitarian assistance.

II. Related Works

Panjaitan et al. in [1] explored the integration of drone technology into conventional paddy field operations to improve agricultural productivity and efficiency. Their study demonstrated that unmanned aerial vehicles (UAVs) can perform multiple tasks such as field monitoring, crop spraying, and data collection with high precision and minimal human intervention. This research emphasizes the significance of automation and adaptability in drone systems, serving as a foundation for developing multi-purpose aerial platforms like M.A.P.A.S, which aims to extend these capabilities across broader operational domains.

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Saha et al. in [2] presented a system utilizing First-Person View (FPV) drones equipped with GPS for surveillance in remote or inaccessible regions. Their research focused on enhancing the reliability and control of FPV systems through advanced communication links and GPS-based navigation. The study highlighted the importance of real-time visual feedback and pilot-assisted maneuverability, which are central design aspects of the M.A.P.A.S project. By integrating FPV control into a larger autonomous framework, M.A.P.A.S extends the application of FPV drones beyond surveillance to multi-mission operations such as rescue and inspection.

He et al. in [3] investigated the use of drones within public safety networks, focusing on data security, communication integrity, and operational reliability. Their work revealed how drones can enhance emergency response and situational awareness in disaster management and law enforcement scenarios. The study also identified vulnerabilities in communication and control systems, suggesting the need for robust and secure integration between autonomous and piloted systems. M.A.P.A.S addresses these aspects by implementing secure communication protocols and coordination between its Mother Drone and Pilot-Assisted Drone for safe and efficient operations.

Singh et al. in [4] analyzed the deployment of drones for contactless food delivery during the COVID-19 pandemic. Their findings demonstrated how UAVs can be scaled to handle logistical challenges, reduce human contact, and operate safely in densely populated or restricted areas. The study validated drones as viable solutions for critical supply delivery, especially under emergency or constrained conditions. The M.A.P.A.S framework draws inspiration from such applications by incorporating modular payload systems for adaptable missions, including medical supply transport, relief distribution, and environmental monitoring.

Xiang et al. in [5] developed a life-ring delivery drone designed for rescue operations in coastal environments. Their system featured precise navigation, high payload efficiency, and rapid deployment capabilities to deliver rescue equipment to drowning victims. This work underscores the importance of accuracy, reliability, and dual-mode (autonomous and manual) control in life-critical missions. The M.A.P.A.S project builds upon these principles by designing a dual-drone architecture—combining a long-range autonomous carrier with a smaller, manually piloted unit for localized precision tasks such as targeted delivery or close-range inspection.

Alqahtani and Aljoufi [6] proposed a YOLOv3-tiny-based UAV system for real-time human detection, employing a PID controller for autonomous tracking. With an mAP of 85.17% and 51 FPS, the model surpasses Faster R-CNN and SSD in processing efficiency, though it remains constrained by battery limitations and performance drops in low-light conditions.

Kumar et al. [7] developed an automated surveillance drone system designed for military applications, incorporating both face recognition and object detection capabilities. The framework employs Histogram of Oriented Gradients (HOG) and Deep CNN embeddings for facial analysis , alongside the YOLOv5 algorithm for real-time identification of weapons and vehicles. Video data is transmitted via GSM to a Python-based server, where a real-time database triggers immediate alerts upon detection. While the system demonstrated accurate detection results during testing, its operational utility is currently constrained by limited flight time and transmitter range.

Singh et al. [8] designed an autonomous drone system capable of real-time surveillance and active fire suppression, utilizing a Raspberry Pi 5 for edge-based processing to eliminate cloud latency dependencies. The architecture integrates a YOLOv8 model with a multi-sensor fusion framework—combining RGB, thermal, and LIDAR data via Kalman filters—to achieve a mean Average Precision (mAP) of 92.6% in hazard detection. Unlike passive monitoring solutions, this platform features an onboard servo-controlled fire extinguisher that automatically activates upon verifying fire threats, offering a cost-effective and responsive solution for industrial safety applications.

Upadhyay et al. [9] designed a smart surveillance quadcopter system utilizing a Raspberry Pi controller to automate security monitoring in restricted areas. The framework employs Haar Cascade classifiers and the Local Binary Patterns Histograms (LBPH) algorithm for precise face recognition to distinguish between authorized personnel and intruders. Additionally, the system integrates the YOLO algorithm for real-time object detection, enabling it to identify potential threats such as weapons and trigger immediate web-based alerts.

Xiao et al. [10] presented a comprehensive survey on vision-based learning for drones, emphasizing its critical role in enhancing autonomy through improved visual perception and decision-making. The study systematically categorizes vision-based control strategies into indirect, semi-direct, and end-to-end approaches, while analyzing applications ranging from single-agent tasks to complex multi-agent and heterogeneous systems. Furthermore, the work explores future directions involving large language models (LLMs) and embodied intelligence, highlighting their potential to advance drones toward Artificial General Intelligence (AGI) in physical 3D environments.

III. Methedology

The proposed M.A.P.A.S system follows a hybrid operational methodology that combines autonomous flight control with human-in-the-loop intervention. The overall system architecture is illustrated in Fig. 1, which presents the functional

block diagram of the coordinated dual-drone platform, while the interaction between system components and users is represented through the use-case diagram in Fig. 2.

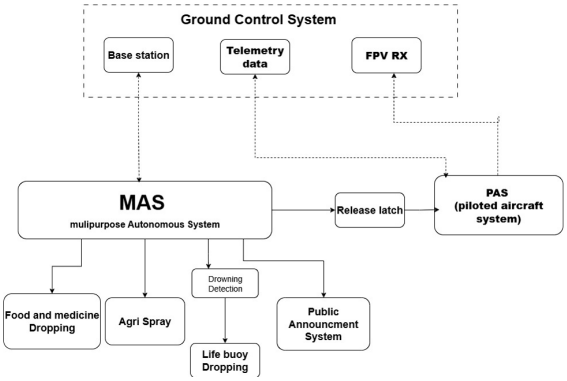

Fig. 1. Block Diagram

As shown in Fig. 1, the Multipurpose Autonomous System (M.A.S) acts as the primary aerial platform responsible for autonomous navigation, mission execution, and payload handling. A flight controller integrated with a companion computing unit enables sensor fusion, real-time perception, and decision-making. Visual data acquired through onboard cameras is processed using computer vision algorithms to detect objects or individuals relevant to the mission. Based on these detections or predefined mission parameters, the system initiates actions such as payload deployment, public announcements, or release of the Piloted Aircraft System (P.A.S).

The P.A.S operates as a precision intervention unit under manual FPV control, allowing accurate maneuvering in confined or high-risk environments. This human-in-the-loop approach ensures reliable execution of tasks that require visual judgment and fine control. A modular payload interface allows rapid reconfiguration of both platforms for different mission objectives.

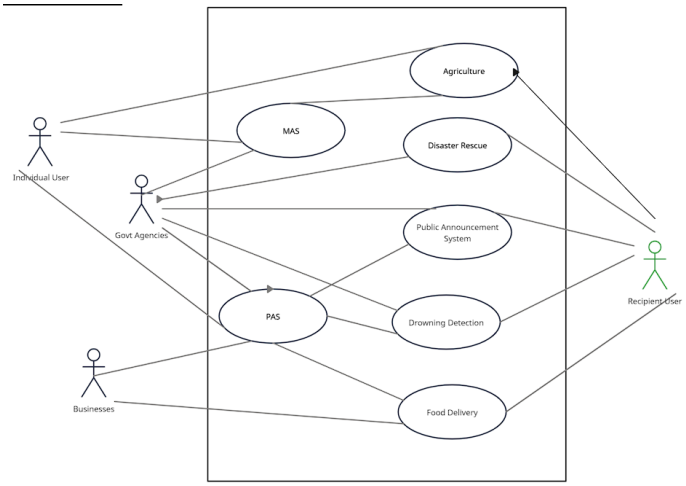

Fig. 2. Use case Diagram

The operational workflow and user interaction are depicted in Fig. 2, where multiple stakeholders such as individual operators, government agencies, and commercial users interact with the system through a ground control station. The coordinated interaction between autonomous operation, piloted control, and user commands enables flexible mission execution and supports diverse real-world applications.

IV. Result and Discussion

The proposed M.A.P.A.S platform is expected to demonstrate reliable coordination between autonomous navigation and piloted precision control within a unified dual-drone framework. The autonomous carrier drone is anticipated to achieve stable long-range flight while executing predefined mission paths and supporting real-time perception using onboard vision algorithms. Successful detection of mission-relevant objects and individuals is expected to trigger timely system responses, including payload deployment and sub-drone release, validating the effectiveness of the perception-to-action pipeline.

Fig. 3. MAS

The piloted FPV drone is expected to provide high maneuverability and low-latency control, enabling accurate operation in confined or high-risk environments where autonomous navigation may be less reliable. This hybrid operational model is expected to improve task completion rates in complex mission scenarios when compared to single-mode UAV systems. The modular payload interface is expected to reduce mission reconfiguration time and support multiple application profiles without major hardware modifications.

From a system-level perspective, the proposed architecture is expected to enhance mission flexibility, reduce deployment

Fig. 4. PAS

cost by minimizing the need for multiple specialized drones, and improve operator safety by limiting direct human exposure to hazardous environments. These outcomes indicate that the M.A.P.A.S framework can serve as a scalable foundation for future multi-role UAV platforms integrating autonomy with human-guided control.

V. Conclusion and Future Scope

This paper presented the Multipurpose Autonomous and Piloted Aircraft System (M.A.P.A.S), a hybrid dual-drone platform designed to overcome the limitations of conventional single-purpose UAVs. By integrating a long-endurance autonomous carrier with a piloted FPV-based sub-drone, the system enables both wide-area monitoring and precise close-range intervention within a unified operational framework. The modular payload architecture and vision-assisted decision-making enhance mission adaptability, reduce response time, and improve operational safety across applications such as disaster response, public safety, agriculture, and humanitarian assistance.

Future work will focus on enhancing system intelligence and scalability through advanced deep learning–based perception, multi-drone coordination, and adaptive mission planning. Improvements in communication reliability using high-bandwidth and low-latency networks, as well as energy-efficient power management, are also planned to extend operational endurance. Additionally, integration with regulatory-compliant autonomous traffic management systems and real-world field validation will further support large-scale deployment of the proposed platform.

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